

ROBOTIC VISION USING MATLAB

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Building a robot is not only a passion but also a dream for most budding engineers. Competitions and R&D in robotics are already being encouraged in various institutions in our country. This article describes a visual sensor system used in the field of robotics for identification and tracing of the object. The program is designed to capture a red ball through a PC-based webcam using Matlab software.

The article does not aim at designing of electronic circuits and mechanical parts of the robot. It describes image capturing and processing techniques, followed by an introduction to actual robotic application to trace the red ball using the serial COM port of the PC.

Robotic vision and control

The whole system of making a robot to follow a red ball can be divided into four blocks: image acquisition, processing of image, decision-making and motion control.

Image acquisition can be achieved by using a PC-based webcam or a digital video camera. This device will capture the image and send it to the camera processor for further processing in the computer. Its main function is to convert the light energy received into electrical signals.

Image processing involves conversion of RGB colour images into grayscale images, setting of threshold levels, saturation of the features into binary images and setting of cut-off values to remove noise in the binary image.

Decision-making is done through the software program and motion control through either software or constant



Fig. 1: Block diagram of the robotic vision and control system

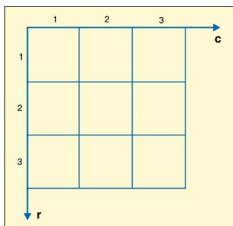


Fig. 2: Pixel coordinates

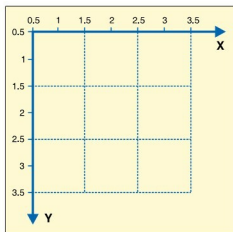


Fig. 3: Spatial coordinates

monitoring by the operator from the keyboard.

Before going to the software program in detail, let us see the coordinate systems to understand the image acquisition process.

Pixel coordinates. Generally, the most convenient method for expressing locations in an image is to use pixel coordinates. Here, the image is treated as a grid of discrete elements, ordered

from top to bottom and left to right. That is, the first component 'r' (the row) increases down-

ward, while the second component 'c' (the column) increases to the right (see Fig. 2). For example, the data for the pixel in the third row, second column is stored in the matrix element (3, 2). When the syntax for a function uses 'r' and 'c,' it refers to the pixel coordinate system. You can use Matlab matrix subscripting to access the values of individual pixels.

Spatial coordinates. Consider a pixel as a square patch. From this perspective, a location such as (3.3, 2.2) is a spatial coordinate. Here, locations in an image on a plane are described in terms of 'x' and 'y.' When the syntax uses 'x' and 'y,' it refers to the spatial coordinate system. Fig. 3 shows the coordinate convention of the spatial coordinate system. Notice that 'y' increases downward.

An image may be defined as a two-dimensional $f(x, y)$ function, where 'x' and 'y' are spatial coordinates, and the amplitude of 'f' at any pair of coordinates (x, y) is called the intensity or gray level of the image at the point. The finite discrete values of the coordinates (x, y) and amplitude of 'f' are the digital images.

An intensity image is a data matrix whose values have been scaled to represent intensities. When the elements of an intensity image are of class 'uint8' or class 'uint16,' they have integer values in the range [0, 255] and [0, 65535], respectively. The class of the elements used in this program is 'uint8.'

Software program

After installing the Matlab software, you will see its icon on your compu-